

HW2.5 — Precision, Bandwidth, and the Cost of Attention

GPU Assignment I · Chelsi Shulamite · M.S. Applied Data Intelligence, San José State University · September 2026

NVIDIA GeForce RTX 5090 · UUID GPU-a3a60fad-ef40-08fa-3eb7-9c22d773bd54 · ISB 840, machine 36

Summary. On this card, dense BF16 matrix multiplication reached 233.8 TFLOPS and an elementwise add reached 1568 GB/s (87.5% of the specified 1792 GB/s). Naive attention memory grows as exactly 4 bytes per L^2 , two FP16 $L \times L$ matrices, while the fused kernel grows only linearly. Under a 20-minute load the card was limited by its 575 W power cap, not by temperature, and held about 97% of its cold-start clock. Three items are incomplete and are stated as such in their sections: the attention OOM boundaries were not reached, the FP8 attempt was not executed, and thermal throughput was measured by a clock proxy.

Part A — Provenance

Field	Value	Source
GPU / UUID	NVIDIA GeForce RTX 5090 / GPU-a3a60fad-ef40-08fa-3eb7-9c22d773bd54	nvidia-smi -q
Driver / CUDA	610.60 / 13.3 (driver); runtime 12.8, torch 2.11.0+cu128	nvidia-smi -q; JSONL records
VRAM	32607 MiB (32 GB), 420 MiB reserved	nvidia-smi -q
Power limit	575 W current/default (range 400–575 W)	nvidia-smi -q
Driver model	Windows WDDM; display attached and active on this card	nvidia-smi -q
Architecture	Blackwell (GB202)	[1]
Memory	32 GB GDDR7, 512-bit, 1792 GB/s	[1], [2]
Tensor cores	5th generation; TF32, FP16, BF16, FP8, FP6, FP4, INT8	[1]
Theoretical peaks (dense)	FP32 104.8 · TF32 104.8 · FP16/BF16 209.5 TFLOPS (FP32 accumulate; 419 with FP16 accumulate), at 2.41 GHz boost	[1]

[1] NVIDIA RTX Blackwell GPU Architecture whitepaper,

images.nvidia.com/aem-dam/Solutions/geforce/blackwell/nvidia-rtx-blackwell-gpu-architecture.pdf, accessed 2026-09-22.

[2] NVIDIA GeForce RTX 5090 product page, [nvidia.com](https://www.nvidia.com), accessed 2026-09-22. PyTorch FP16/BF16 matmuls accumulate in FP32, so 209.5 TFLOPS is the relevant peak.

Part B — Precision and achieved throughput

torch.mm, $N \times N$, 5 warm-up iterations, CUDA-event timing, 10 repetitions for $N < 8192$ and 5 for $N \geq 8192$. The benchmark was invoked twice; tables report the mean.

N	FP32	TF32	FP16	BF16
1024	43.90 (41.9%)	71.20 (67.9%)	115.69 (55.2%)	87.97 (42.0%)
4096	68.77 (65.6%)	91.31 (87.1%)	205.98 (98.3%)	178.16 (85.0%)
8192	66.75 (63.7%)	111.48 (106.4%)	223.78 (106.8%)	229.53 (109.6%)
16384	66.38 (63.3%)	115.64 (110.3%)	216.03 (103.1%)	228.80 (109.2%)

Achieved TFLOPS (% of theoretical peak).

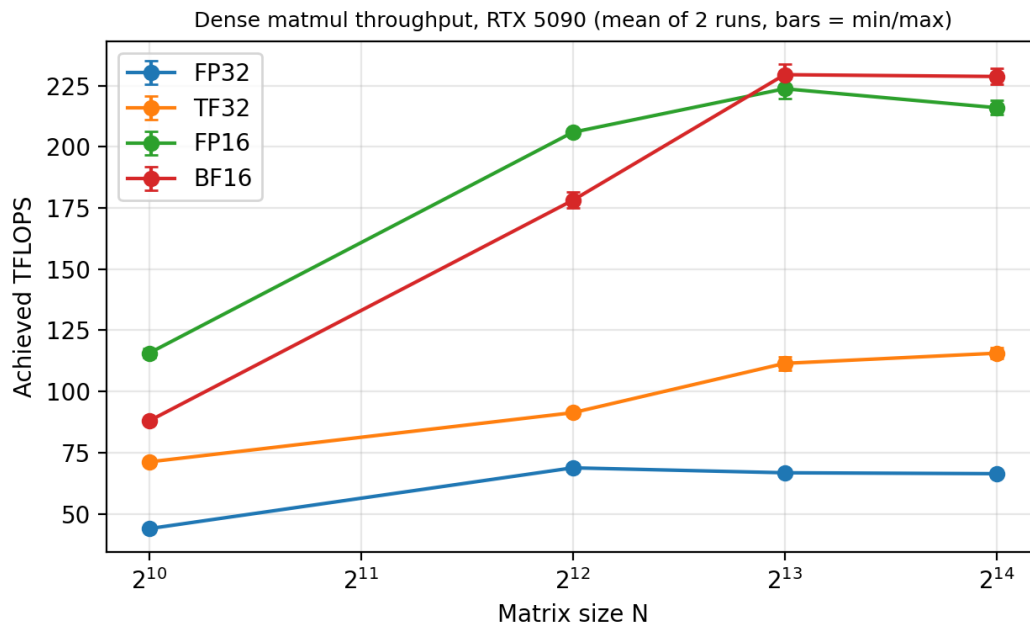


Figure 1. Achieved TFLOPS vs. matrix size, one line per precision. Replotted from results/precision.jsonl as the mean of the two invocations (the committed precision_tflops.png plots both invocations as one connected line).

Plateaus. FP32 plateaus by $N = 4096$; FP16 and BF16 by $N = 8192$; TF32 is near plateau at $8192\text{--}16384$ (+3.7% over that range, after +22% from 4096 to 8192).

Why small matrices never plateau. At $N = 1024$ a matmul takes only $\sim 18\text{--}50\ \mu\text{s}$, so fixed launch overhead is a large share of the runtime. The problem is also too small to fill the GPU: with a typical 128×128 output tile, a 1024×1024 result has only 64 tiles for 170 SMs, leaving most SMs idle. At 4096 there are 1024 tiles, about six waves, and the remaining loss is wave quantization.

Above 100% of peak. The whitepaper peaks assume a 2.41 GHz boost clock; this card ran at ~ 2.53 GHz sustained and up to ~ 2.60 GHz when cold, 5–8% higher. That explains most of the excess. At 2.60 GHz the BF16 peak would be about 226 TFLOPS, so the best run (233.8) is still $\sim 3\%$ over, within the up-to-4% spread between the two invocations caused by the low repetition count. Numbers are reported as measured, not capped.

FP8. Not executed. An attempt using `torch._scaled_mm` with `float8_e4m3fn` was prepared but not run within the reserved time, so there is no result or error to report. Preparing it showed that the FP8 path is less mature than `torch.mm`: it is a private API that requires explicit scale tensors and a column-major second operand.

Part C — Bandwidth-bound vs. compute-bound

Operation	Measured (mean; best)	Arithmetic intensity	Roofline side
Elementwise add, 256M FP32	1526.3 GB/s (85.2%); 1568.3 GB/s (87.5%) of 1792	1 FLOP / 12 B = 0.083 FLOP/B	Memory-bound
8192×8192 FP16 matmul	223.9 TFLOPS; 229.8 TFLOPS	$2N^3 / (3N^2 \cdot 2\text{ B}) = 2731$ FLOP/B	Compute-bound

Ridge point = $209.5\text{ TFLOPS} \div 1792\text{ GB/s} \approx 117\text{ FLOP/byte}$. The add sits three orders of magnitude below it and reaches 85–88% of peak bandwidth while using a negligible fraction of compute. The matmul sits $23\times$ above it and is limited by arithmetic.

Part D — The cost of attention

Single head, **head_dim = 128**, **batch = 1**, FP16. Naive: $\text{softmax}(QK^T \cdot \text{scale}) \cdot V$, materializing the full $L \times L$ matrix. Fused: `scaled_dot_product_attention` (backend selected automatically by PyTorch). 1 warm-up, 3 repetitions, two invocations averaged.

L	Naive (ms)	Fused (ms)	Speedup	Naive peak (MiB)	Fused peak (MiB)
512	0.230	0.090	2.56×	9.62	8.62
1024	0.111	0.091	1.22×	13.12	9.12
2048	0.259	0.151	1.71×	26.12	10.12
4096	0.178	0.273	0.65×	76.12	12.12
8192	0.632	0.515	1.23×	272.12	16.12
16384	2.301	1.992	1.16×	1048.12	24.12

Latency caveat. The harness generated Q, K and V inside the timed region and took only 3 repetitions, so sub-millisecond rows, including the 4096 row where fused appears slower, are dominated by overhead and noise. Memory measurements are unaffected: they were identical in both invocations.

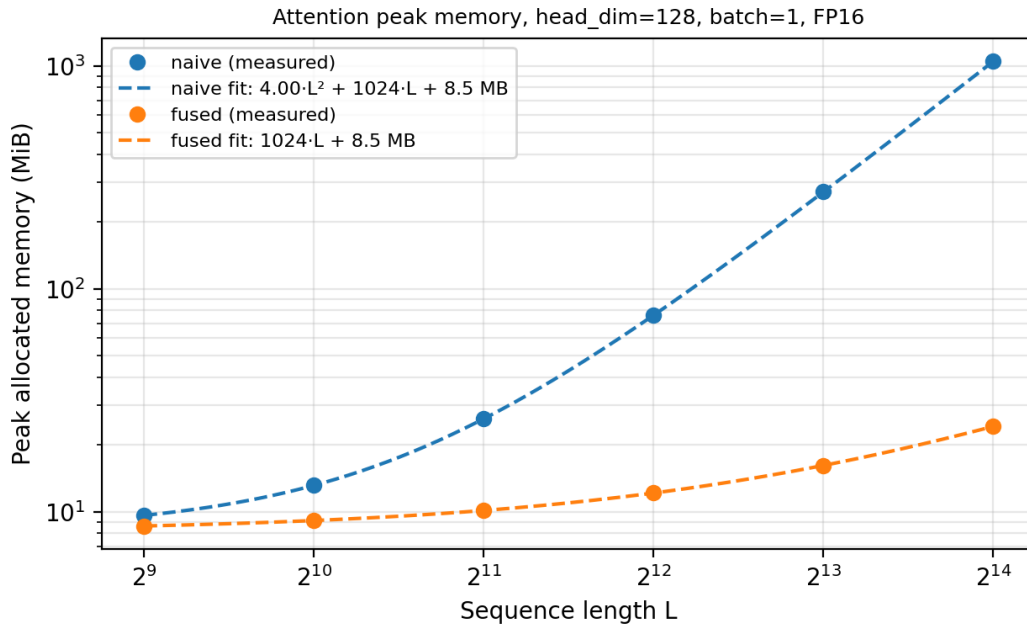


Figure 2. Peak memory vs. sequence length (log-log) with fitted curves. Replotted from results/attention.jsonl; fits computed on naive and fused rows separately.

Fit. Least squares on my own data:

```
naive: peak_bytes = 4.00 * L^2 + 1024 * L + 8.52 * 10^6 (R^2 ≈ 1.0)
fused: peak_bytes = 1024 * L + 8.52 * 10^6 (no L^2 term)
```

Measured quadratic coefficient: 4.0 bytes per L^2 , equal to two FP16 $L \times L$ matrices alive at peak: the scaled scores and the softmax output. The shared linear term, 1024 bytes per token, is Q, K, V and the output at head_dim 128 in FP16 ($4 \times 128 \times 2$ bytes). *Correction:* the committed results/attention_fit.txt reports $2.057 \cdot L^2$ because the harness fitted naive and fused rows together; the naive-only fit above supersedes it.

OOM boundaries — not reached. Both implementations succeeded at every tested length, so the largest success is 16384 for both, and there is no measured smallest failure. As a *projection only*, the naive fit predicts OOM near $L \approx 90,000$ on ~ 31 GiB of usable memory; the fused path at 1024 bytes/token would not OOM until ~ 30 million tokens. The extended sweep and bisection needed to measure the boundaries were not run within the reserved time.

What the fused kernel avoids. The naive path writes the full $L \times L$ score matrix to GPU memory, reads it back for the softmax, writes the result out again, and reads it once more for the second matmul, so both its memory and its memory traffic scale with L^2 . The fused kernel loads Q , K and V in blocks that fit in on-chip SRAM, computes the softmax incrementally with a running maximum and normalizer, and accumulates the output tile by tile. The $L \times L$ matrix therefore never exists in global memory, which is exactly what the fits show: identical linear terms, and the quadratic term present only in the naive path.

Part E — Sustained load and thermal behaviour

8192×8192 FP16 matmul in a continuous loop for 1200 s, sampled every 5 s (results/thermal.csv, 241 samples of GPU clock, memory clock, temperature, power and utilization).

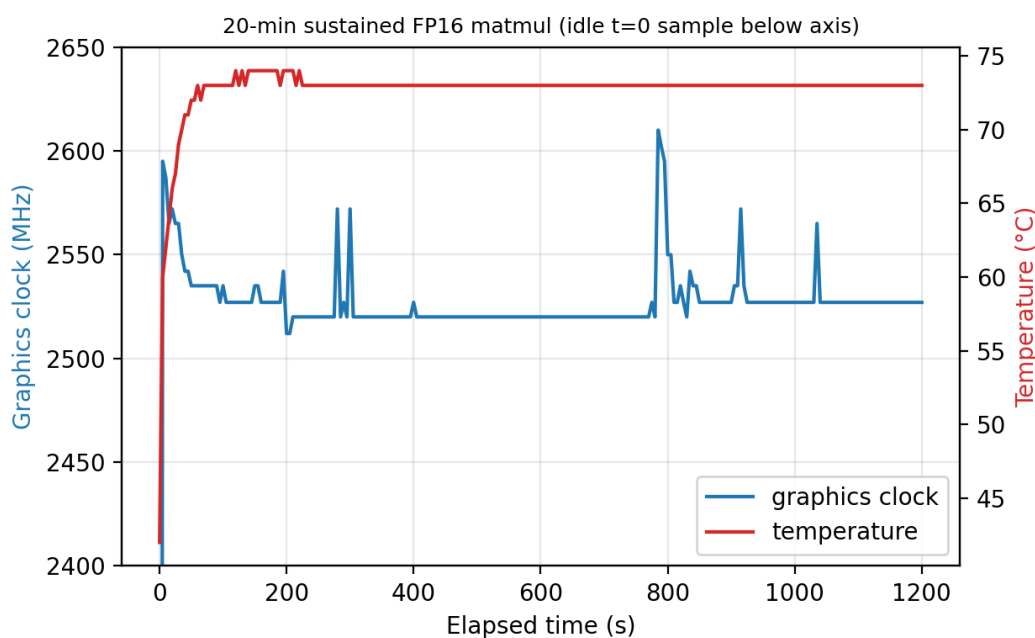


Figure 3. Graphics clock and temperature vs. time. Replotted from results/thermal.csv with a legend and a zoomed clock axis; the idle first sample (1132 MHz, 42 °C at $t = 0.16$ s) is below the axis range.

- **Power** hit the 575 W limit at the first loaded sample ($t = 5$ s) and stayed at 572–585 W throughout.
- **Clock** peaked at 2595 MHz at $t = 5$ s and settled to 2520–2535 MHz by $t \approx 50$ –80 s, with brief single-sample spikes (to 2610 MHz near $t \approx 780$ s).
- **Temperature** rose from 60 °C to a 73–74 °C plateau by $t \approx 70$ s. The memory clock was fixed at 13801 MHz, and utilization was 100%.

Throttling. There was no thermal throttling: the temperature plateaued at 73–74 °C. The card was **power-limited from ~ 5 s**, and the $\sim 2.5\%$ clock decline over the first minute is the boost algorithm trading frequency for power as the die warmed. Throttle-reason flags were not logged, so this classification rests on the power and temperature traces.

Peak vs. steady state. Per-sample throughput was not logged, so graphics clock serves as a proxy; that is valid for a fixed-size, compute-bound matmul at 100% utilization. Peak clock in the first 30 s was 2595 MHz, and the mean over the final 5 minutes was 2528.8 MHz, giving **steady-state / peak = 97.4%** (98.2% against the first-30 s mean).

Part F — Table HW2.5.1

Measurement	RTX 5090 (GPU-a3a60fad-ef40-08fa-3eb7-9c22d773bd54)	Notes
Peak achieved TFLOPS (BF16)	233.77	N = 8192, best run, precision.jsonl
% of theoretical peak (BF16)	111.6%	vs. 209.5 TFLOPS [1]; above-boost clocks, Part B
Effective bandwidth (GB/s)	1526.3 (best 1568.3)	85.2% (87.5%) of 1792, roofline.jsonl
Naive attention OOM length	Not reached; largest success 16384	Projected ~90k from the 4.0 B/L ² fit
Fused attention OOM length	Not reached; largest success 16384	Linear, 1024 B/token
Steady-state / peak throughput	97.4% (clock proxy)	thermal.csv
Throttle onset (s, or none)	No thermal throttle; power-limited from ~5 s	575 W cap; max 74 °C

Interpretation

The two numbers that bound the rest of this course differ by three orders of magnitude: this card performs about 117 FLOPs in the time it moves one byte, so any operation below that intensity waits on memory rather than arithmetic. The elementwise add is a pure bandwidth test and large matmuls are pure compute tests. Precision matters only on the compute side, where FP16/BF16 tensor cores deliver about 3.4× FP32 throughput once N is large enough to fill the SMs. Attention shows the stakes: naive memory of 4 bytes per L² is only 1 GiB at L = 16384 for one head and one sequence, but it multiplies by batch × heads × layers in a real model, while the fused kernel does the same computation in linear memory. Sustained performance on this card is set by its power limit rather than temperature and costs only 2–3% relative to a cold start.

Limitations

- Low repetition counts (5–10 per matmul, 3 per attention configuration) and no per-configuration variance recorded; the two invocations differ by up to ~4%.
- Attention latencies include input generation inside the timed region.
- The benchmark script did not receive the SID-derived seed (it ran as a subprocess); the reported timing and memory results do not materially depend on tensor values.
- Run times in RUN_LOG.txt are reconstructed from file timestamps, not logged live.
- The card was also driving a display under the Windows WDDM driver during all measurements.